

- Description

This module manages a smart treadmill system, ensuring that:

1. The treadmill continuously monitors and updates its current speed (in km/h) and incline (in percentage).
2. As the user runs, the system records the distance covered and calculates the calories burned using a numerical formula.
3. The total distance and total calories burned are updated based solely on numerical inputs.

This system helps users track their exercise performance and calorie expenditure in real

- Specification

module SmartTreadmillUsage

var currentSpeed, incline, totalDistance, totalCalories: real

process updateSpeed(newSpeed: real)

functional scenario form:

newSpeed > 0 && Speed updated successfully.

end process

process updateIncline(newIncline: real)

functional scenario form:

(newIncline ≥ 0) && Incline updated successfully.

end process

process addDistance(distance: real) totalCalories: real

functional scenario form:

(distance > 0) && totalDistance = totalDistance + distance, totalCalories = totalCalories + distance * (1.2 * currentSpeed + 0.5 * incline), Distance added and calories updated.

end process

end module

- Java Code

```
package code;
class SmartTreadmill {
    private double currentSpeed;
    private double incline;
    private double totalDistance;
    private double totalCaloriesBurned;
    public SmartTreadmill(double initialSpeed, double initialIncline) {
        this.currentSpeed = initialSpeed;
        this.incline = initialIncline;
        this.totalDistance = 0.0;
        this.totalCaloriesBurned = 0.0;
    }
}
```

```

    }
    public void updateSpeed(double newSpeed) {
        currentSpeed = newSpeed;
        System.out.printf("Current speed updated to: %.2f km/h\n", currentSpeed);
    }
    public void updateIncline(double newIncline) {
        incline = newIncline;
        System.out.printf("Current incline updated to: %.2f%%\n", incline);
    }
    public void addDistance(double distance) {
        if (distance <= 0) {
            System.out.println("Distance must be positive.");
            return;
        }
        totalDistance += distance;
        double calories = distance * (1.2 * currentSpeed + 0.5 * incline);
        totalCaloriesBurned += calories;
        System.out.printf("Added %.2f km. Total Distance: %.2f km, Calories Burned: %.2f cal\n", distance, totalDistance, totalCaloriesBurned);
    }
    public double getTotalCaloriesBurned() {
        return totalCaloriesBurned;
    }
}

public class c71_SmartTreadmillSystem {
    public static void main(String[] args) {
        SmartTreadmill treadmill = new SmartTreadmill(8.0, 5.0);

        // testcase-VT:
        // treadmill.addDistance(1.0);
        // treadmill.updateSpeed(10.0);
        // treadmill.addDistance(2.0);
        // treadmill.updateIncline(10.0);
        // treadmill.addDistance(1.5);
        // treadmill.addDistance(-1.5);

        // testcase-FT:
        // treadmill.addDistance(1.7976931348623157E308);
        // treadmill.addDistance(1.9490628022799995E291);
        // treadmill.addDistance(6.361740986641919E293);
    }
}

```

```
//      treadmill.addDistance(1.8287080229312227E302);
//      treadmill.addDistance(4.1845584833267585E298);
//      treadmill.addDistance(2.9235942034199993E289);

// testcase-Z3:
//      treadmill.addDistance(1.0);
//      treadmill.updateSpeed(10.0);
//      treadmill.addDistance(2.0);
//      treadmill.updateIncline(10.0);
//      treadmill.addDistance(1.5);
//      treadmill.addDistance(-1.5);

// testcase-UVT:
//      treadmill.addDistance(1.0);
//      treadmill.updateSpeed(10.0);
//      treadmill.addDistance(2.0);
//      treadmill.updateIncline(10.0);
//      treadmill.addDistance(1.5);
//      treadmill.addDistance(-1.5);
}
}
```

- Coverage report

1. Test case number: 6
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.2 %	2	840	842
✓ src	0.2 %	2	840	842
✓ code	0.2 %	2	840	842
✓ c71_SmartTreadmillSystem.java	100.0 %	2	0	2
> c71_SmartTreadmillSystem		0	0	0
> SmartTreadmill	100.0 %	2	0	2

3. Vibration testing instruction coverage: 95.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.6 %	128	20,788	20,916
✓ src	0.6 %	128	20,788	20,916
✓ code	0.6 %	128	20,788	20,916
✓ c71_SmartTreadmillSystem.java	95.5 %	128	6	134
> c71_SmartTreadmillSystem	89.3 %	25	3	28
> SmartTreadmill	97.2 %	103	3	106

4. Fuzz testing branch coverage: 50%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.1 %	1	841	842
✓ src	0.1 %	1	841	842
✓ code	0.1 %	1	841	842
✓ c71_SmartTreadmillSystem.java	50.0 %	1	1	2
> c71_SmartTreadmillSystem		0	0	0
> SmartTreadmill	50.0 %	1	1	2

5. Fuzz testing instruction coverage: 71.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	96	20,820	20,916
▼ src	0.5 %	96	20,820	20,916
▼ code	0.5 %	96	20,820	20,916
▼ c71_SmartTreadmillSystem.java	71.6 %	96	38	134
> c71_SmartTreadmillSystem	89.3 %	25	3	28
> SmartTreadmill	67.0 %	71	35	106

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.2 %	2	840	842
▼ src	0.2 %	2	840	842
▼ code	0.2 %	2	840	842
▼ c71_SmartTreadmillSystem.java	100.0 %	2	0	2
> c71_SmartTreadmillSystem		0	0	0
> SmartTreadmill	100.0 %	2	0	2

7. Z3 instruction coverage: 95.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.6 %	128	20,788	20,916
▼ src	0.6 %	128	20,788	20,916
▼ code	0.6 %	128	20,788	20,916
▼ c71_SmartTreadmillSystem.java	95.5 %	128	6	134
> c71_SmartTreadmillSystem	89.3 %	25	3	28
> SmartTreadmill	97.2 %	103	3	106

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.2 %	2	840	842
▼ src	0.2 %	2	840	842
▼ code	0.2 %	2	840	842
▼ c71_SmartTreadmillSystem.java	100.0 %	2	0	2
> c71_SmartTreadmillSystem		0	0	0
> SmartTreadmill	100.0 %	2	0	2

9. Unimproved V-method instruction coverage: 95.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.6 %	128	20,788	20,916
▼ src	0.6 %	128	20,788	20,916
▼ code	0.6 %	128	20,788	20,916
▼ c71_SmartTreadmillSystem.java	95.5 %	128	6	134
> c71_SmartTreadmillSystem	89.3 %	25	3	28
> SmartTreadmill	97.2 %	103	3	106